

1) Explain About Windows CE?

* Windows CE later known as Windows Embedded CE and Windows Embedded Compact, is a discontinued operating system developed by Microsoft for mobile and embedded devices.

* Originally targeted solely at handheld computers, Windows CE evolved into a component-based, embedded, realtime operating system.

* It was part of the Windows Embedded family and served as the software foundation of several products including the Pocket PC, AutoPC, Windows Mobile, Windows Phone 7 and others.

* Windows CE can be configured as Monolithic kernel or hybrid kernel, dependent upon the target device type and the manufacturer.

* Earlier versions of Windows CE worked on MIPS and SuperH architectures, but in version 7.0 released in 2011 when the product was also renamed to Embedded Compact support for these were dropped but remained for MIPS II architecture.

Features :

* Windows CE is optimized for devices that have minimal memory; a Windows CE kernel may run with one megabyte of memory.

* Devices are often configured without disk storage, and may be configured as a closed system that does not allow for end-user extension.

* Windows CE conforms to the definition of a real-time operating system, with a deterministic

interrupt latency.

- * The first version known during development under the code name "Pegasus" featured a Windows like GUI and a number of Microsoft's popular apps, all trimmed down for smaller storage, memory and speed of handheld devices.

- * Many platform have been based on the core Windows CE operating system including Microsoft's AutoPC, PocketPC 2000, Pocket PC 2002 etc.

Development Tools:

- * The embedded Microsoft Visual C++ - a tool for development of embedded apps for Windows CE. It can be used standalone using the SDK exported from Platform Builder or using the Platform Builder's platform Manager connectivity setup.

- * GCC project provides GNU development tools, such as GCC, GCC C++ and binutils that target Windows CE.

- * CodeGear Delphi Prism - runs in Visual Studio also supports the .NET Compact Framework and thus can be used to develop mobile apps.

- * It employs the Oxygen compiler created by RemObjects software, which targets .NET, the .NET Compact Framework, and Mono.

- * Basic4PPC - a programming language similar to embedded Visual Basic, targets the .NET Compact Framework and supports Windows CE and Windows Mobile devices.

- * GIL Basic - a very easy to learn and use Basic dialect that compiles for many platforms.

Windows CE - AS Technical.

- * Windows CE is a 32-bit multitasking, multi-thread pocket version of Windows for mobile devices.

- * Like the regular Windows OS, Windows CE also used the same Win32 Programming interface.

- * Windows CE could run on less than 1MB of RAM, keeping with its primary use case: small footprint mobile and embedded devices.

- * Windows CE could run on less than 1MB of RAM, keeping with its primary use case small footprint it also run on many different processor architectures including SH3, MIPS and ARM.

- * Windows CE supported almost all Windows Technologies, including .NET compact Framework, MFC, COM, ActiveX and DirectX.

- * It also came preinstalled as firmware on the ROM chips built into portable devices from various vendors.

- * Users could synchronize the files on their Windows CE mobile devices with Windows-powered desktop PCs using an RS-232 serial cable or infrared connection, later using network based syncing.

- * Numerous components of Windows CE are available in its source code through the shared source initiative.

- * The code can be modified to work with specific platforms.

2) Explain About Brew?

* Binary Runtime environment for wireless (BREW also known as Brew MP or Qualcomm BREW) is an obsolete application development platform created by Qualcomm, originally for code division multiple access (CDMA) mobile phones, featuring third-party applications such as mobile games.

* It was offered in some feature phones (mostly with specifications similar to those of mid to high end mobile phones) as well as smartphones.

* First developed in 1999 as a platform for wireless applications on CDMA-based mobile phones, it debuted in September 2001.

* As a software platform that can download and run small programs for playing games, sending messages, and sharing photos, the main advantages of Brew MP was that the application developers could easily port their applications among all Brew MP device by providing a standardized set of application programming interfaces.

* The BREW runtime library is part of the wireless device onchip firmware or operating system to allow programmers to develop applications without needing to code for system interface or understand wireless applications.

* BREW was not a virtual machine such as Java ME, as it runs code natively.
Working principles:

* BREW works by providing a standardized runtime environment on mobile devices.

* Developers write applications using C/C++ with the help of BREW SDK Tools.

* These applications are then tested using emulators and must go through a certification process.

* Once approved, they are distributed through mobile carriers, and users can download them over the air (OTA).

* The BREW runtime environment on the device executes these applications and manages their interaction with hardware components.

Architecture:

* The BREW architecture is divided into multiple layers. At the bottom is the hardware layer, which includes processor, memory and communication modules.

* Above it is the operating system layer, which controls basic device operations.

* The BREW runtime environment layer sits above the OS and provides APIs and services for application execution.

* The top layer is the application layer, where user applications run.

Supporting components include the BREW SDK, API libraries and application Manager, which handles installation, updates and billing.

Development Process:

- * The BREW application development process involve several steps.
- * First, developers design the application and write code using BREW APIs in C/C++.
- * Next, the application is tested using BREW emulator tools.
- * After testing, the application must be digitally signed and submitted for carrier certification.
- * Once approved, it is deployed through the carrier network, allowing users to download and install it on their devices.

Key Features:

- * BREW offers a range of features that made it significant in early mobile computing.
 - * It provides platform independence, allowing applications to run on different devices with minimal modification.
 - * It ensures high performance due to low-level programming.
 - * Security is enforced through digital signatures and strict certification processes.
 - * BREW supports OTA downloads, enabling direct installation of apps from the network.
- Services provided by BREW.
- * These includes user interface service for display and input handling, multimedia services for audio and video playback, messaging services such as SMS integration, network services for

internet connectivity, location services using GPS and file system access for data storage and retrieval.

Advantages:

- * It includes efficient resource usage, strong security mechanisms, reliable performance and controlled distribution through carriers.
- * It is well-suited for devices with limited memory and processing power.

Disadvantages:

- * It is mainly designed for feature phones and lacks support for modern smartphone capabilities.
- * Development is complex due to the use of C/C++ and strict certification requirements.
- * The dependence on mobile carriers restricts application distribution.
- * Compared to modern platforms, limited graphical user interface capabilities and fewer development tools.